

Healthcare Payment Variance Analytics and Contract Modeling Simulator

Using Real CMS Medicare FY2022 Inpatient Data and FY2025 IPPS Rate Schedules

Peter Chika Ozo-ogueji

po3783a@american.edu | M.S. Data Science, M.S. Business Analytics & AI

American University, Washington, D.C.

ORCID: 0009-0003-7898-4644

<https://github.com/PeterOzo/healthcare-payment-variance-analytics>

Abstract

We present a production-grade payment variance analytics system applied to real CMS Medicare data to identify and quantify underpayment across the U.S. hospital system. Our system ingests five publicly available federal datasets — spanning 145,879 hospital-DRG combinations, 4.95 million Medicare discharges, and 2,906 hospitals — and applies the FY2025 IPPS expected payment formula with three critical IFC-corrected parameters to compute variance between expected and actual Medicare payments. We identify \$16.66 billion in total underpayment exposure, with 88.5% of all analyzed hospital-DRG pairs exhibiting statistically significant underpayment. The national average yield ratio is 0.818, indicating hospitals receive 81.8 cents for every dollar of expected Medicare reimbursement. We further develop a contract renegotiation scenario modeling framework that evaluates four distinct contract parameter changes — rate escalation, stop-loss threshold reduction, denial rate optimization, and high-weight DRG carve-outs — projecting net revenue impacts ranging from \$137.3 billion (carve-out scenario) to \$184.2 billion (stop-loss reduction scenario) across the analyzed discharge portfolio. Our dual-threshold variance classification framework, false-variance suppression logic, and DuckDB-based SQL analytical engine constitute original methodological contributions over naive groupby approaches common in the literature. All code, data, and analytical marts are reproducible from publicly available CMS sources without proprietary data access.

1. Introduction

Healthcare payment variance — the gap between the expected reimbursement for a clinical service and the actual payment received — represents one of the most analytically tractable and financially significant problems in hospital revenue cycle management. The American Hospital Association (2024) estimates combined Medicare and Medicaid underpayments exceed \$100 billion annually, with Medicare alone paying approximately \$0.84 per dollar of hospital cost. Despite this scale, the dominant industry practice remains retrospective: variance reports are generated after claim adjudication, flagged discrepancies are triaged manually, and appeals are filed within shrinking timely-filing windows. This reactive workflow is structurally disadvantaged relative to payer-side payment integrity systems, which increasingly apply machine learning to pre-adjudication claim editing.

The problem is simultaneously difficult and important for several reasons. First, the expected payment formula under the Inpatient Prospective Payment System (IPPS) involves hospital-specific adjustments — wage index, indirect medical education (IME), and disproportionate share (DSH) factors — that vary across 3,159 provider CCN codes and must be joined correctly to produce accurate expected payments. Second, the distinction between true underpayments and false variances (arising from lesser-of provisions, transfer adjustments, or charge audit activity) requires multi-factor classification logic. Third, CMS data files use encoding schemes and multi-row header structures that require robust ETL engineering. Fourth, scaling to 145,879 hospital-DRG pairs demands efficient analytical SQL rather than row-level Python operations.

This project addresses these challenges through a complete end-to-end pipeline: ingestion with encoding fallback, Pandera schema validation, vectorized IPPS expected payment computation, DuckDB SQL

analytics with window functions and CTEs, contract scenario modeling, and Power BI-ready export. Our key contributions are: (1) a correctly implemented FY2025 IPPS formula using the IFC-corrected base rate of \$6,624.39; (2) a dual-threshold variance classification framework with false-variance suppression; (3) a parameterized contract scenario model evaluating four distinct renegotiation strategies; and (4) a complete set of analytical marts and interactive visualizations derived entirely from public CMS data.

2. Related Work

The academic and industry literature on healthcare payment analytics spans three related streams: prospective payment system methodology, revenue cycle analytics, and contract modeling.

Prospective Payment System Methodology. The IPPS framework, established under the Tax Equity and Fiscal Responsibility Act of 1982 and implemented in 1983, has been extensively analyzed in health economics. Newhouse and Byrne (1988) documented the behavioral responses of hospitals to DRG-based payment, and subsequent work by Sloan et al. (1988) and Cutler (1995) examined the distributional effects of wage-index adjustments. The MedPAC Payment Basics series (2024) provides the authoritative current reference for IPPS, OPSS, and physician fee schedule methodology. Critically, the October 2024 Interim Final Rule with Comment (CMS-1808-IFC, 89 FR 80452) — arising from the D.C. Circuit's *Bridgeport Hospital v. Becerra* ruling — corrected the FY2025 IPPS base rate from \$6,394.36 to \$6,624.39, a correction absent from most practitioner literature.

Revenue Cycle Analytics. Ensemble Health Partners (2022) documented that approximately 70% of underpayments lie beyond the standard payment variance report — outside the claims that trigger automated alerts — suggesting systematic underdetection in current tooling. Pi et al. (2024) analyzed the relationship between Case-Mix Index and hospitalization costs across DRG groups, establishing the CMI benchmark of approximately 1.79 for Medicare hospitals that we use as a national reference. The HFMA MAP Keys initiative (hfma.org, 2024) standardizes 29 revenue cycle KPIs including Remittance Denial Rate and Denial Write-Offs that anchor our scenario modeling outputs. Rivet Health (2024) estimates that 2.5-3% of all billed claim lines are underpaid across payer types.

Contract Modeling and Payment Variance Detection. Practical Data Solutions (2023) and MD Clarity (2024) describe commercial approaches to payer contract modeling, emphasizing the reanalysis of historical utilization under proposed contract parameters. Aspirion (2024) reports a 35% underpayment reduction and 0.25-1% net revenue improvement through systematic variance recovery programs. Prior academic work on payment variance detection is sparse; most published analyses focus on aggregate Medicare underpayment rather than hospital-DRG level variance with false-variance suppression. Our work extends this literature by providing a reproducible, open-data framework with explicit lesser-of and stop-loss suppression logic.

3. Approach

3.1 FY2025 IPPS Expected Payment Formula

The expected Medicare operating payment for a given hospital-DRG combination is computed as:

$$P_{\text{expected}} = [(\lambda \times WI) + (1 - \lambda)] \times B \times W_{\text{DRG}} \times (1 + IME + DSH) \times 0.98$$

where λ is the labor share (0.676 for hospitals with wage index $WI > 1.0$; 0.620 for $WI \leq 1.0$ per the statutory floor), $B = \$6,624.39$ is the FY2025 IFC-corrected national standardized amount, W_{DRG} is the MS-DRG relative weighting factor from Table 5 (Correction Notice file), IME is the indirect medical education adjustment derived from the resident-to-beds ratio r as $IME = 1.35 \times ((1 + r)^{0.405} - 1)$, DSH is the empirical disproportionate share factor from the Impact File, and 0.98 is the sequestration factor.

Three implementation decisions are critical to correctness. First, the IME and DSH adjustments are applied additively as $(1 + IME + DSH)$, not multiplicatively as $(1 + IME) \times (1 + DSH)$. The multiplicative form inflates expected payments by $IME \times DSH$ at high-DSH teaching hospitals, a compounding error of 3-5%

in magnitude. Second, the FY2025 base rate $B = \$6,624.39$ reflects the IFC correction of October 2024; using the original Final Rule rate of $\$6,394.36$ understates expected payments by $\$229.53$ per discharge, a $\$1.14$ billion aggregate error across the full dataset. Third, all CCN and DRG codes are maintained as zero-padded strings (CCN: 6 characters; DRG: 3 characters). Integer casting strips leading zeros and silently corrupts every join to the Impact File.

3.2 Variance Classification Framework

For each hospital-DRG pair we compute: total expected payment $E = P_{\text{expected}} \times N$ (where N is total discharges), total actual Medicare payment $A = \text{avg_medicare_pymt} \times N$, variance $V = E - A$, variance percentage $\text{Pct} = V / E \times 100$, and yield ratio $Y = \text{avg_medicare_pymt} / P_{\text{expected}}$.

A dual-threshold rule classifies underpayment: a pair is flagged as underpaid if and only if $|\text{Pct}| > 5\%$ AND $|V| > \$200$. This prevents small-volume DRGs with large percentage variances from dominating the signal. Five classification categories are produced:

- True Underpayment: $V > 0$, dual threshold satisfied, lesser-of flag absent. Yield $Y < 0.92$ triggers high-priority escalation.
- Overpayment: $V < 0$, dual threshold satisfied. Documented for compliance review.
- False Variance — Lesser-Of: $\text{avg_billed} < P_{\text{expected}}$ AND $\text{avg_medicare_pymt} \geq \text{avg_billed} \times 0.95$. Payer correctly applied lesser-of provision; suppressed from underpayment count.
- Stop-Loss Outlier: $\text{avg_billed} > \$85,000$. Separate outlier payment analysis applied.
- Compliant: All other pairs within dual-threshold tolerance.

3.3 ETL Pipeline Architecture

The ingestion layer implements a four-encoding fallback cascade ($\text{cp1252} \rightarrow \text{latin-1} \rightarrow \text{utf-8-sig} \rightarrow \text{utf-8}$) to handle CMS Windows-1252 formatted files. CCN and DRG codes are normalized at load time. Each DataFrame carries `_source` and `_dq_flags` audit columns for downstream traceability. Pandera schema validation gates are applied at every layer boundary, asserting null rates, numeric ranges, and key format compliance.

The analytical layer uses DuckDB in-memory SQL for `RANK() OVER PARTITION BY` state and MDC ranking queries, payer yield benchmarking CTEs, Case-Mix Index computation, and stop-loss proximity detection. DuckDB executes the full 145,879-row analytical suite in under 5 seconds on Colab T4 hardware, versus approximately 50 seconds for equivalent pandas groupby chains.

3.4 Contract Scenario Modeling

Four contract renegotiation scenarios are modeled by re-pricing historical discharge volume under modified contract parameters:

- S1 — Rate Escalation: Commercial rate increased from 190% to 215% of Medicare.
- S2 — Stop-Loss Reduction: Threshold reduced from $\$100\text{K}$ to $\$75\text{K}$; payout rate increased from 60% to 65% of excess charges.
- S3 — Denial Rate Halved: Managed care denial rate reduced from 8% to 4% via front-end eligibility and prior authorization improvement.
- S4 — High-Weight DRG Carve-Out: DRGs with relative weight > 3.0 repriced at 125% of Medicare; all others remain at 190%.

For each scenario s , the net revenue per discharge under the new contract is computed as $R_s = (P_{\text{expected}} \times \text{base_rate}_s + \text{stop_loss_add}_s) \times (1 - \text{denial_rate}_s)$, where base_rate_s incorporates carve-out logic where applicable. Total portfolio uplift is the aggregate of $(R_s - \text{avg_medicare_pymt}) \times N$ across all pairs with valid expected payments.

4. Experiments

4.1 Data

Five publicly available federal datasets constitute the analytical corpus, all sourced from cms.gov or data.cms.gov without data use agreements:

Dataset	Source	Key Stats	Role
CMS Medicare Inpatient FY2022	data.cms.gov	145,879 rows; 2,906 CCNs; 540 DRGs	Primary: actual payment data
IPPS Table 5 FY2025 (CN file)	cms.gov IPPS FR	771 MS-DRGs; weight 0.547–37.71	DRG relative weights
IPPS Impact File FY2025 (IFC)	cms.gov CMS-1808-IFC	3,159 hospitals; 62 provider variables	Wage index, IME, DSH per CCN
CMS Outpatient Provider CY2022	data.cms.gov	APC-level payment data	Outpatient OPPS layer
CY2025 Physician Fee Schedule RVU25A	cms.gov PFS FR	10,000+ HCPCS; CF = \$32.35/RVU	Part B benchmark reference

The primary dataset spans 4.95 million total discharges across 51 U.S. states and territories, with a discharge-weighted national Case-Mix Index of 1.8448 — above the typical community hospital benchmark of 1.40–1.80, consistent with the complex-patient composition of the CMS utilization file.

4.2 Evaluation Method

We evaluate the payment variance framework on four dimensions. First, formula correctness: expected payment computations for DRGs with published benchmark values (e.g., DRG 470 Major Hip/Knee Replacement) are verified against CMS illustrative examples. Second, classification precision: the dual-threshold rule is compared against a naive single-threshold (percentage-only) baseline to quantify false-positive suppression from lesser-of logic. Third, geographic and service-line coverage: variance estimates are aggregated at state and MDC level to assess distributional completeness. Fourth, scenario financial impact: net revenue uplift per scenario is validated by scaling per-hospital estimates (at ~15,000 discharges) against HFMA and Ensemble industry benchmarks.

The primary metric is total underpayment exposure (USD), defined as the sum of positive variances across all flagged hospital-DRG pairs after false-variance suppression. Secondary metrics include national average yield ratio Y_{nat} , proportion of pairs with underpayment flags, high-priority account count ($Y < 0.92$), and stop-loss outlier exposure.

4.3 Experimental Details

All computations are implemented in Python 3.10 (pandas 2.x, NumPy 1.26) with DuckDB 1.1.0 for SQL analytics and Pandera 0.20 for schema validation. Experiments were run on Google Colab with T4 GPU (16GB VRAM; 12GB RAM). The full pipeline — ingestion, validation, IPPS computation, SQL analytics, scenario modeling, and export — completes in approximately 4 minutes on this hardware. The vectorized IPPS formula processes all 145,879 rows in a single NumPy operation without loops or apply() calls. DuckDB registers all DataFrames as zero-copy virtual tables, eliminating serialization overhead. Parquet exports use snappy compression; Excel workbooks use openpyxl with six analytical mart sheets.

4.4 Results

Table 2 presents the primary quantitative results of the payment variance analysis:

Metric	Value
Total hospital × DRG pairs analyzed	145,879
Total Medicare discharges	4,950,000+
Unique hospitals (CCN)	2,906
Unique MS-DRGs represented	540
National Case-Mix Index (discharge-weighted)	1.8448
Underpayment flags (dual-threshold)	129,138 (88.5%)
Total underpayment exposure	\$16.66 billion
Total overpayment detected	\$1.86 billion
National avg. yield ratio Y_nat	0.818 (81.8 cents / dollar expected)
High-priority accounts (Y < 0.92)	125,502
Stop-loss outlier accounts (billed > \$85K)	51,411 (\$7.80B variance)
Lesser-of flags suppressed	Documented; not double-counted

Table 3 presents the top DRGs by national underpayment exposure:

DRG	Description	Underpayment	Avg Yield	# Hospitals
871	Septicemia w/o MV (MCC)	\$2.016B	79.3%	1,847
291	Heart Failure and Shock (MCC)	\$720.9M	78.7%	2,234
853	Infectious Diseases w/ O.R. Procedure	\$671.6M	78.1%	1,102
177	Respiratory Infections (MCC)	\$369.9M	80.5%	1,923
870	Septicemia with MV >96 Hours	\$320.6M	76.8%	891

Contract scenario modeling results are presented in Table 4. All figures are portfolio-level aggregates across 4.95 million discharges; individual hospital impact scales proportionally to discharge volume and CMI.

Scenario	Label	Parameter Changed	Uplift vs. Actual	CMI
S1	Rate Escalation	Commercial rate: 190% → 215% of Medicare	\$174.3B	1.8612
S2*	Stop-Loss Reduction (BEST)	Threshold: \$100K → \$75K; payout 65%	\$184.2B	1.8612
S3	Denial Rate Halved	MA denial rate: 8% → 4% (operational)	\$163.8B	1.8612
S4	High-Weight DRG Carve-Out	DRGs with weight > 3.0 at 125% Medicare	\$137.3B	1.8612

* S2 recommended as the primary negotiation anchor, consistent with the \$7.80B stop-loss variance finding (Section 4.4).

5. Analysis

Several qualitative and quantitative findings emerge beyond the topline numbers.

Geographic concentration vs. yield severity. California leads in absolute underpayment exposure (\$2.43B) due to discharge volume, but its average yield ratio of 75.4% is among the lowest nationally — suggesting structural payer behavior, not simply scale. States with smaller absolute exposure often have worse yield ratios, indicating that geographic targeting by discharge volume alone misses systematic payer non-compliance patterns.

Service line clustering. MDC 05 (Cardiovascular) and MDC 18 (Infectious Disease) account for \$7.39B — 44.4% of total underpayment — despite representing a minority of DRG count. This clustering reflects payer audit concentration: high-CMI DRGs attract disproportionate charge audit activity, and septicemia/cardiovascular cases carry both high billed charges and high coding complexity that creates audit vulnerability.

Stop-loss concentration effect. 51,411 stop-loss accounts (\$85K+ billed) carry \$7.80B in variance — 46.8% of total underpayment from 35.2% of flagged pairs. This super-linear concentration is consistent with the charge audit mechanism: payers target stop-loss threshold accounts to suppress total billed charges below the outlier threshold, eliminating the additional outlier payment. Accounts within 10% of the threshold (85% of charge floor) showed the highest variance-per-account in the dataset.

DRG-level critical yield pattern. 25 DRGs exhibited average yield ratios below 85% (CRITICAL tier). The highest-volume critical account — DRG 310 (Cardiac Arrhythmia Without CC/MCC) — had a 69.5% yield across 24,508 discharges. This is a low-complexity, high-volume DRG that should not attract aggressive payer audit activity, suggesting a systematic contract interpretation error rather than charge-audit-induced suppression.

Scenario S2 analytical justification. The stop-loss threshold reduction scenario (S2) outperforms the simple rate escalation scenario (S1) by \$9.9B despite a lower base rate improvement, because it directly addresses the \$7.80B stop-loss variance concentration. This counterintuitive result — that threshold restructuring outperforms percentage-point rate increases — is analytically consistent with the finding that 46.8% of underpayment is concentrated in accounts where payers have contractual incentive to suppress outlier payments.

Scenario S3 operational significance. The denial rate halving scenario (\$163.8B uplift) requires no contract renegotiation — it is achievable through front-end eligibility verification and prior authorization compliance. This positions S3 as the highest-ROI, fastest-implementation path: no payer consent required, implementation begins in the current contract term. HFMA estimates 85% of denials are avoidable through automation, suggesting $\$163.8B \times 0.85 = \$139.2B$ of the projected uplift is operationally accessible.

False variance suppression impact. The lesser-of suppression mechanism removes a meaningful number of apparent underpayments attributable to cases where billed charges fall below the contracted rate. Without this suppression, the underpayment flag count would be inflated and recovery resources would be directed at non-recoverable accounts. The dual-threshold rule additionally prevents sub-\$200 variances from generating work queues with negligible recovery potential.

6. Conclusion

This project presents a production-grade healthcare payment variance analytics system that identifies \$16.66 billion in Medicare underpayment exposure across 145,879 hospital-DRG combinations using only publicly available CMS data. The primary methodological contributions — IFC-corrected IPPS formula

implementation, dual-threshold variance classification with false-variance suppression, and parameterized contract scenario modeling — collectively address limitations of the ad-hoc variance reporting tools commonly deployed in hospital revenue cycle departments.

The analysis reveals that underpayment is structurally concentrated in high-complexity service lines (cardiovascular and infectious disease), stop-loss outlier accounts, and specific high-volume low-complexity DRGs suggesting systematic contract interpretation errors. The stop-loss threshold reduction scenario is identified as the highest-value contract renegotiation target, generating \$184.2 billion in projected portfolio-level uplift.

The primary limitation of this work is the use of FY2022 actual payment data as the utilization base for FY2025 rate scenarios. Volume mix and DRG weight shifts between 2022 and 2025 may alter scenario projections at the hospital level. A second limitation is the absence of real commercial payer contract terms; scenarios are parameterized by percent-of-Medicare multipliers rather than actual payer-specific contract language.

Future directions include: integration of the NY SPARCS multi-payer dataset (2.4 million discharges with commercial, Medicaid, and Medicare payer flags) to enable genuine multi-payer yield benchmarking; development of a prospective ML risk scorer that flags claims at risk of underpayment prior to adjudication using BiLSTM or XGBoost on HCPCS/DRG feature vectors; and a Power BI live-connection dashboard enabling real-time contract performance monitoring.

References

- [1] American Hospital Association. (2024). Costs of Caring: Trends in Hospital Care Costs, 2024. AHA. <https://www.aha.org/costsofcaring>
- [2] American Hospital Association. (2024). Fact Sheet: Underpayment by Medicare and Medicaid. AHA. <https://www.aha.org/fact-sheets/2020-01-07-fact-sheet-underpayment-medicare-and-medicare>
- [3] Aspirion. (2024). Payment Variance Recovery. Aspirion RCM Services. <https://www.aspirion.com/services/payment-variance-recovery/>
- [4] Centers for Medicare & Medicaid Services. (2024). FY2025 IPPS Final Rule — Table 5: MS-DRG Relative Weighting Factors (Correction Notice). cms.gov.
- [5] Centers for Medicare & Medicaid Services. (2024). FY2025 IPPS Interim Final Rule with Comment (CMS-1808-IFC). 89 FR 80452, October 3, 2024. federalregister.gov.
- [6] Centers for Medicare & Medicaid Services. (2023). Medicare Inpatient Hospitals by Provider and Service (FY2022). data.cms.gov.
- [7] Cutler, D. M. (1995). The incidence of adverse medical outcomes under prospective payment. *Econometrica*, 63(1), 29-50.
- [8] Ensemble Health Partners. (2022). Beyond the Payment Variance Report: Resolving Common, Costly Underpayment Issues. Becker's Hospital Review Webinar.
- [9] HFMA. (2024). MAP Keys: Industry-Standard Revenue Cycle KPIs. hfma.org/data-and-insights/map-initiative/map-keys/
- [10] HFMA. (2024). Managing and Measuring Payer Performance. hfma.org/payment-reimbursement-and-managed-care/contracting/54833/
- [11] MD Clarity. (2024). Healthcare Underpayments: A Guide for Providers. mdclarity.com/blog/healthcare-underpayments
- [12] MedPAC. (2024). Hospital Acute Inpatient Services Payment System: Payment Basics. medpac.gov/document-type/payment-basic/
- [13] Newhouse, J. P., & Byrne, D. J. (1988). Did Medicare's prospective payment system cause length of stay to fall? *Journal of Health Economics*, 7(4), 413-416.
- [14] Pi, M., et al. (2024). Using Case Mix Index within DRGs to Evaluate Variation in Hospitalization Costs. PMC10785921. PubMed Central.
- [15] Practical Data Solutions. (2023). Reanalyzing Payer Contracts in Today's Healthcare Landscape. YouTube: youtube.com/watch?v=xTBWsjGgR04

- [16] Rivet Health. (2024). The Ins and Outs of Payment Variance. rivethealth.com/blog/the-ins-and-outs-of-payment-variance
- [17] Sloan, F. A., Morrissey, M. A., & Valvona, J. (1988). Medicare prospective payment and the use of medical technologies in hospitals. *Medical Care*, 26(9), 837-853.